Hemang Joshi

Automation & Robotics Engineer — Machine Design, Controls & Embedded

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SUMMARY

Electrical engineer who designs and ships **complete automation machines** — mechanics (**SolidWorks**), power electronics, embedded firmware and the control software on top. Seven production machines built end-to-end and sold to paying customers, including a **3-axis CNC selective soldering robot** and a **100–500 wires/min cutter**. Author of **karmyogi**, a browser-based CAD/CAM and machine-control workbench for 3-axis GRBL machines. **Expert Python** for machine control, instrument control and data acquisition.

SKILLS

Mechanical Design / CAD	**SolidWorks (extensive)** — part & assembly modelling, weldments, sheet metal, GD&T, tolerance stack-up, drawings for manufacture; design for CNC machining, laser cutting & 3D printing; motion & mechanism design
Industrial Controls / PLC	**Siemens PLC & TIA Portal (extensive)**, **Coolmay PLC / HMI** — PLC programming, ladder logic, machine sequencing & interlocks, integrated PLC+HMI panels, drive & I/O integration, commissioning and on-machine debug; built **tiaportal-mcp** — an MCP server / VS Code extension that drives Siemens TIA Portal projects from an AI assistant
AI / Machine Learning	**PyTorch** — transformer & deep-learning models (**TrendMaster**, 276★ on GitHub); **LLM applications**: **RAG** pipelines, embeddings/vector search, agent tooling and **MCP servers** (built **tiaportal-mcp** to drive Siemens TIA Portal from an AI assistant); **machine vision** — inspection & image-based part detection; applying ML to machine data (sensor streams, anomaly & quality detection)
Simulation & Tools	MATLAB / Simulink, instrument control & DAQ, oscilloscope / analyzer bring-up
Embedded & CNC Firmware	STM32 / ESP32 / Arduino firmware, **CNC firmware development (GRBL)** — motion control, step/dir drivers, limits & homing, G-code execution; sensor & transmitter loops, serial / Modbus, real-time control
Software & Web	**Flutter (Android)** — cross-platform mobile apps; **web development with Vite** (JavaScript / TypeScript front-ends), browser-based machine-control UIs — see **karmyogi**
Languages	Python (expert), C / C++, JavaScript / TypeScript, SQL · Git, **Linux**
Communication	Technical-note & specification writing · fluent English · willing to travel

EXPERIENCE

Gadget Access Pty Ltd
Jul 2025 – Present

AI / Automation Engineer — Sydney, Australia · Remote

- Build **Python automation tooling** for internal systems — scripting repetitive workflows and integrating them with the team's existing tools.
- Develop in-house tooling and repeatable automation pipelines.

hjLabs.in
2017 – Present

Founder & Principal Engineer — Automation & Machine Systems — Gandhinagar, India

- Designed and delivered **7+ production machines and automation systems end-to-end** — mechanics (**SolidWorks**), power electronics, embedded firmware, and **Python control software** — shipped to paying customers.
- Wrote **Python** for instrument control and data acquisition (4–20 mA loops, sensors) on the machines he built.

Tata Communications Limited
2020 – 2021

Network Associate — India

- Enterprise network operations and infrastructure support — monitoring and resolving connectivity incidents for business clients.

66 kV Substation, Devla (Dhari)
2019 – 2020

Switchboard Operator — HV Power Systems — Gujarat, India

- Operated and monitored **66 kV switchgear and protection**; hands-on HV power-systems experience with fault response and strict **EHS / safety compliance**.

JETPACE Technologies
2016 – 2019

Embedded Design Engineer — India

- Embedded systems and instrumentation firmware — microcontrollers, sensor interfaces and serial/field communication.

Rotomotive Powerdrives India Ltd
2015 – 2016

Design Engineer — Power Electronics & Drives — Gujarat, India

- Developed power-electronics test setups including a **Testing Panel for an 87 Hz VFD (variable-frequency drive)** and a **DC-motor test-bench**; established the company **R&D testing lab**.
- Designed three-phase (63-frame) and single-phase induction motors for clients (Truetzschler, Wilo, Mather & Platt), validating performance against specification.

SELECTED PROJECTS — TEST, POWER & AUTOMATION

karmyogi — Browser-based dockable **CAD/CAM + machine-control workbench** for 3-axis GRBL machines (CNC, soldering, PCB routing, pen-plotting) — real-time control software bridging operator UI and hardware. Live demo: karmyogi.hjlabs.in · code: github.com/hemangjoshi37a/karmyogi.

87 Hz VFD-Drive Test Panel — Designed an automated test panel to validate variable-frequency-drive (power-converter) operation across the frequency range, with instrumented measurement and reporting.

DC-Motor Test-Bench — Instrumented bench for motor performance characterization (speed/torque/current), with data logging.

4–20 mA Humidity & Temperature Transmitter (loop-powered) — Designed and calibrated an industrial instrumentation loop — directly relevant to test-signal acquisition.

OPEN-SOURCE CODE — GITHUB (60 original repos · github.com/hemangjoshi37a)

tiaportal-mcp — C# MCP server / VS Code extension: AI-assisted Siemens TIA Portal programming.

SensorSync-IoT — ESP32 0–5 V analog data logger (Wi-Fi/MQTT/SD).

Razorpay-Dynamic-QR-HMI-POS — C++ embedded HMI/POS with live UPI QR generation.

AIComputerInteractionLogger ★13 — Python HCI capture & logging for datasets.

claude-code-frontend-dev ★29 — Python automated visual UI-testing tool.

AutoCut — Arduino automatic wire-cutting machine (C++).

TrendMaster ★276 — Transformer stock-price predictor (PyTorch).

AUTOMATION MACHINES — DESIGNED, BUILT & SHIPPED (HJLABS.IN)

Commercial automation machines I built end-to-end — power electronics, embedded firmware & Python control software — each with a live demo video and spec sheet (hjlabs.in):

IIoT Transformer Winding Machine — IoT multi-layer winding control. ▶ demo

AutoSolder 6 — Soldering Robot — programmable PCB soldering. ▶ demo

Wire & Cable Cut + Strip Machine — cut/strip/crimp 0.1–10 mm². ▶ demo

Semi-Auto Spring Coiling Machine — adjustable pitch/dia/length. ▶ demo

IoT DC Motor Speed Controller — PWM drive + web UI.

AutoCut V2 — Wire Cutter — 100–500 wires/min. ▶ demo

SensorSync IoT Data Logger — Wi-Fi/MQTT 0–5 V logging. ▶ demo

EDUCATION & TRAINING

Gujarat Technological University — G. H. Patel College of Engineering & Technology — 2015
B.E., Electrical Engineering — power systems & power electronics — Gujarat, India

Adani Power — Mundra Thermal Power Plant — Industrial Training
Power-generation & HV systems exposure (one of Asia's largest thermal plants) — Gujarat, India

CERTIFICATIONS

Machine Learning — Stanford University / DeepLearning.AI (Andrew Ng) · verify
Introduction & Programming with IoT Boards — POSTECH · verify
Blockchain Basics — University at Buffalo (SUNY) · verify
NCC 'C' Certificate — Ministry of Defence, India

ADDITIONAL

Work authorization: Indian citizen; requires a Singapore Employment Pass (employer-sponsored) · **Mobility:** Ready to relocate to Singapore · **Notice:** Available at 30 days